Chapter 1: Algorithm Analysis and Asymptotic Notation

Asymptotic analysis evaluates the efficiency of algorithms as the input size n grows toward infinity.

Big-O notation represents an asymptotic upper bound. Formally, f(n) = O(g(n)) if there exist positive constants c and n0 such that 0 <= f(n) <= c * g(n) for all n >= n0.

Common complexity classes include O(1) constant time, O(log n) logarithmic time, O(n) linear time, O(n log n) linearithmic time, and O(n^2) quadratic time.

In computer science education, understanding asymptotic behavior is fundamental for selecting optimal data structures such as balanced binary search trees and hash tables.

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Chapter 2: Sorting Algorithms and Divide-and-Conquer

Divide-and-conquer is an algorithmic paradigm that recursively breaks a problem into smaller subproblems of the same type until they become simple enough to solve directly.

MergeSort is an asymptotically optimal comparison-based sorting algorithm. Its recurrence relation is T(n) = 2T(n/2) + O(n), yielding a guaranteed worst-case time complexity of O(n log n).

QuickSort selects a pivot element and partitions the array into elements less than and greater than the pivot. Its average-case time complexity is O(n log n), while its worst-case complexity is O(n^2) when poor pivots are chosen.

In terms of space complexity, MergeSort requires O(n) auxiliary space for merging subarrays, whereas in-place QuickSort requires O(log n) space for recursive call stack frames.

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Chapter 3: Graph Algorithms and Shortest Paths

A graph G = (V, E) consists of a set of vertices V and edges E connecting pairs of vertices.

Dijkstra's algorithm computes the single-source shortest path for a directed or undirected graph with non-negative edge weights.

Using a min-priority heap, Dijkstra's algorithm achieves a time complexity of O((V + E) log V). If a Fibonacci heap is utilized, the running time decreases to O(E + V log V).

The Bellman-Ford algorithm can handle graphs containing negative edge weights and is capable of detecting negative weight cycles in O(V * E) time.

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Chapter 4: Cryptographic Foundations and RSA Algorithm

Asymmetric cryptography, or public-key cryptography, utilizes mathematically linked key pairs consisting of a public key and a private key.

The RSA algorithm was invented in 1977 by Ron Rivest, Adi Shamir, and Leonard Adleman. Its security relies on the computational difficulty of factoring the product of two large prime numbers.

Key generation involves choosing two distinct prime numbers p and q, computing the modulus n = p * q, and Euler's totient function phi(n) = (p - 1) * (q - 1).

A public exponent e is chosen such that 1 < e < phi(n) and gcd(e, phi(n)) = 1. The private decryption key d satisfies the congruence d * e = 1 (mod phi(n)).

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Chapter 5: Dynamic Programming and Optimal Substructure

Dynamic programming is an algorithmic technique for solving optimization problems by decomposing them into overlapping subproblems and optimal substructures.

The two primary implementations of dynamic programming are top-down memoization (recursive caching) and bottom-up tabulation (iterative table construction).

Classic examples include the 0/1 Knapsack problem, Longest Common Subsequence (LCS), and Matrix Chain Multiplication.

By storing the solutions to subproblems rather than recomputing them exponentially, dynamic programming typically reduces polynomial time complexities from O(2^n) to O(n^2) or O(n * W).

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